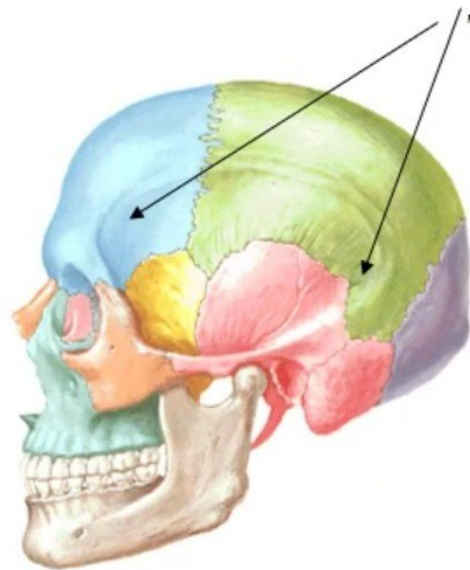


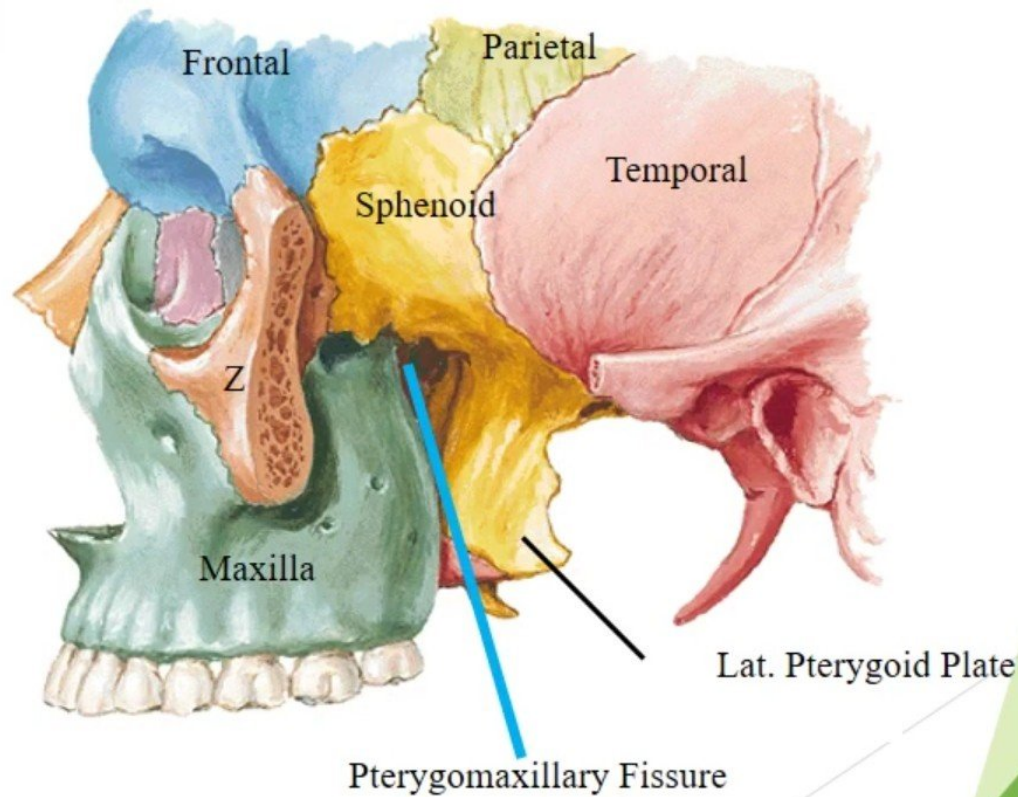
TEMPORAL AND INFRATEMPORAL FOSSAE

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Boundaries of the Temporal and Infratemporal Fossae



Temporal Fossa



Frontal

Parietal

Sphenoid

Temporal

Z

Maxilla

Lat. Pterygoid Plate

Pterygomaxillary Fissure

Infratemporal Fossa

Boundaries:

Lateral=Ramus of Mandible

Anterior=Maxilla

Medial=Lat. Pterygoid Plate

Roof=Sphenoid

Contents

1. Muscles of mastication.
2. Maxillary artery and pterygoid venous plexus.
3. Mandibular and maxillary nerves.

Muscles of Mastication

- ▶ **External Group**

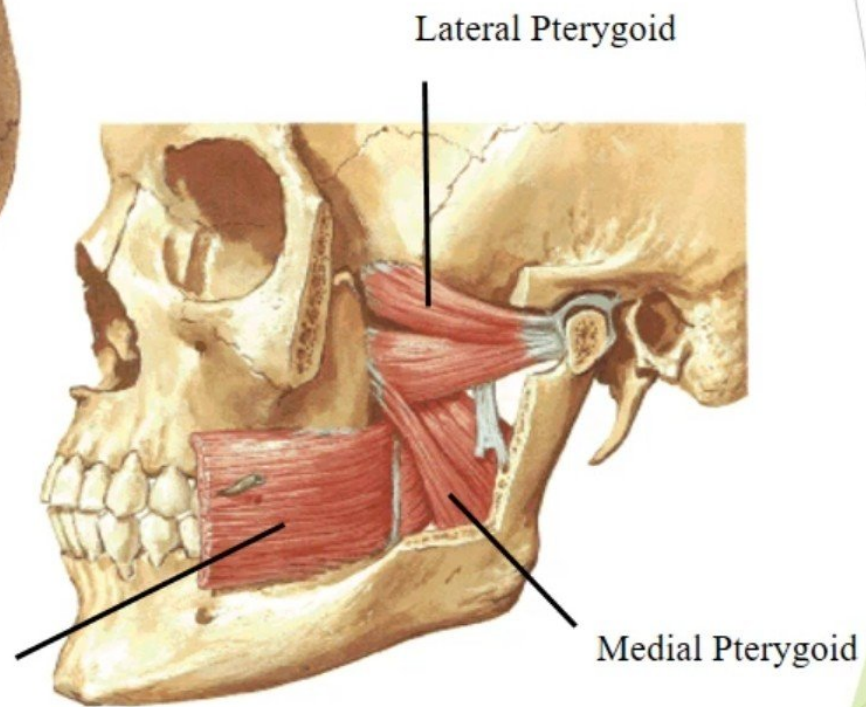
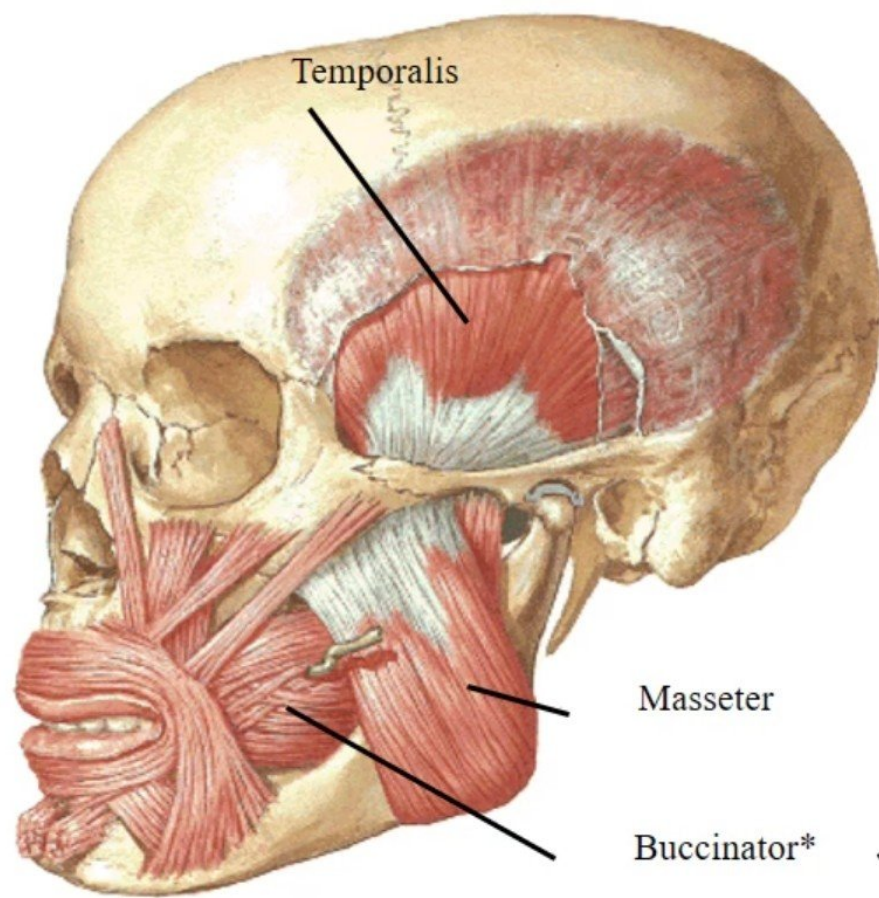
- ▶ Temporalis.

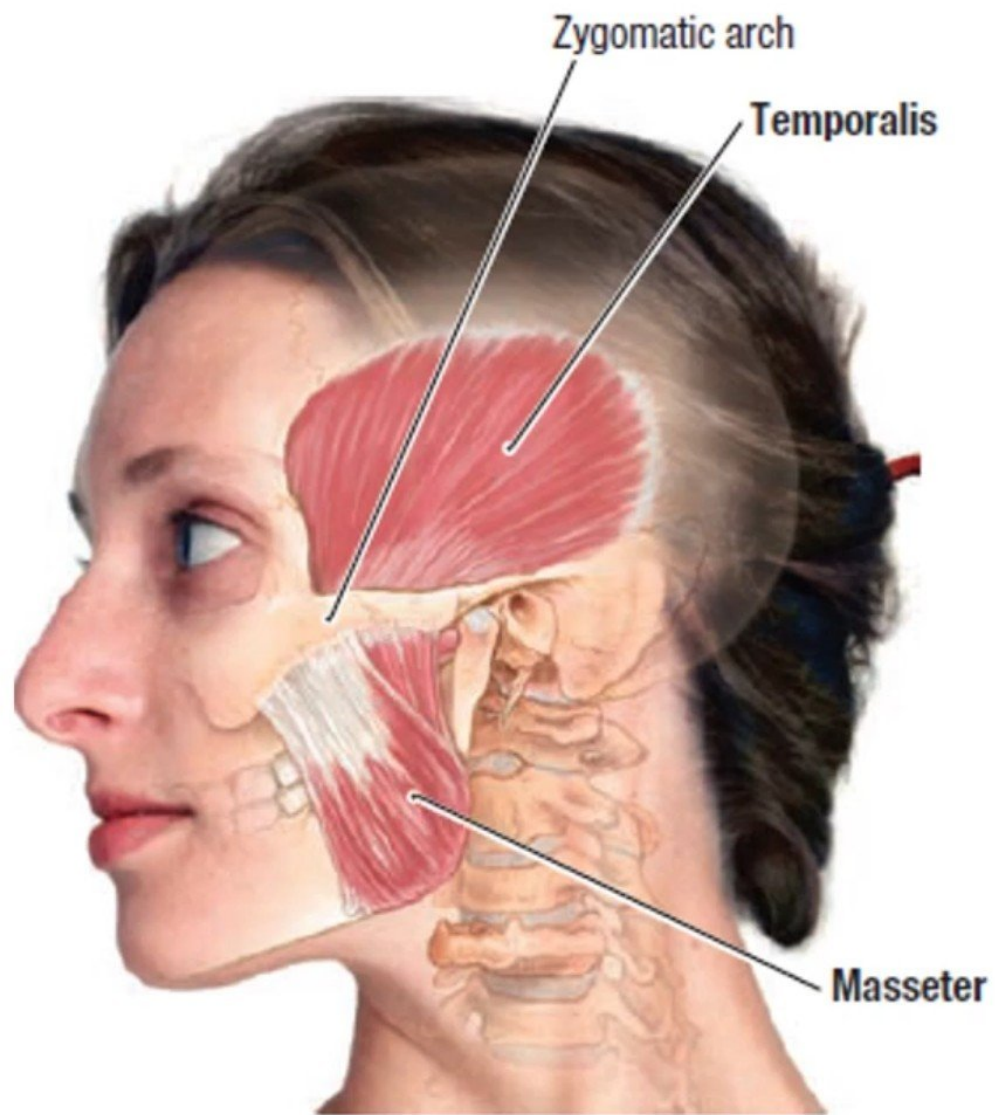
- ▶ Masseter.

- ▶ **Internal Group**

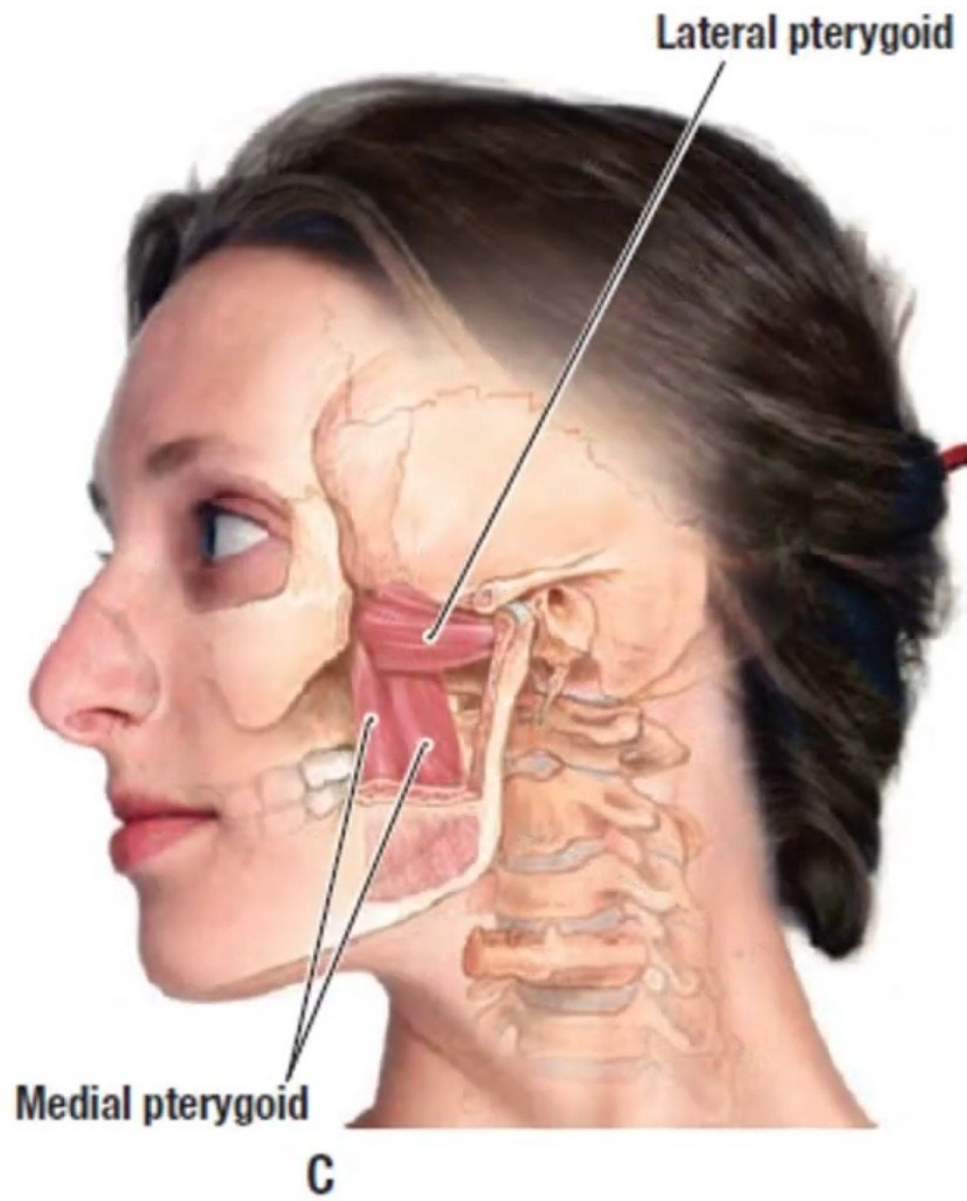
- ▶ Medial pterygoid.

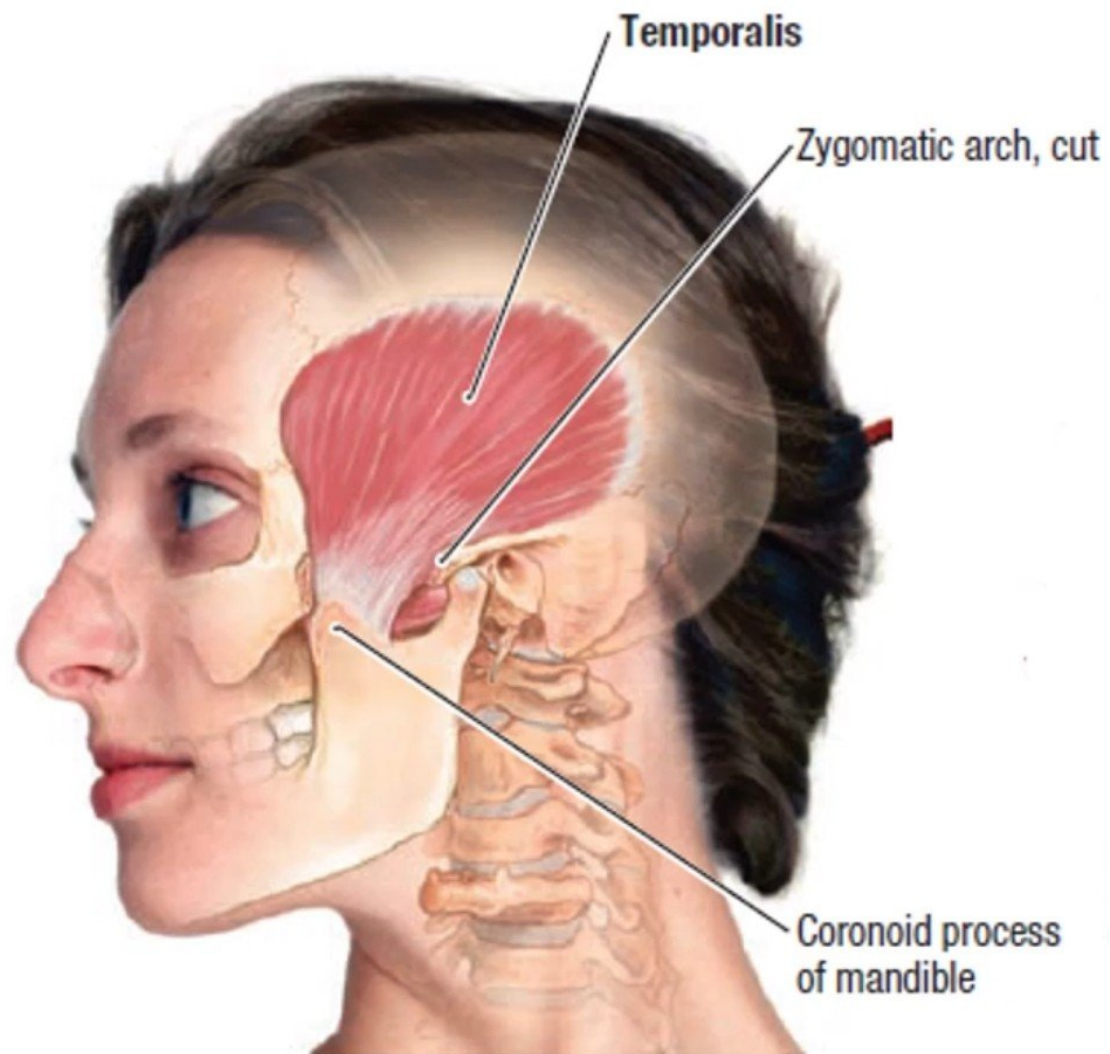
- ▶ Lateral pterygoid.





A



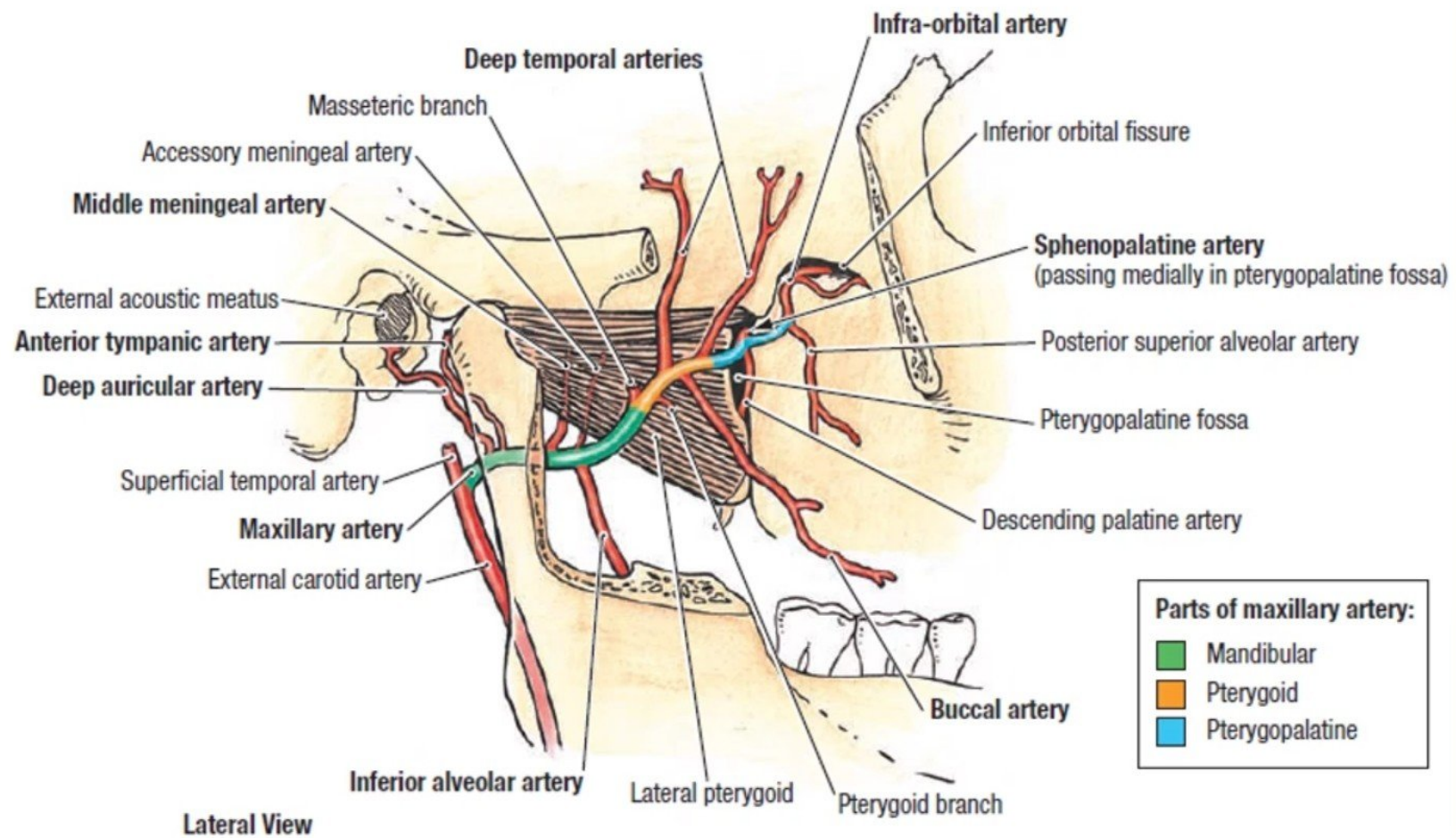


B

Lateral Views

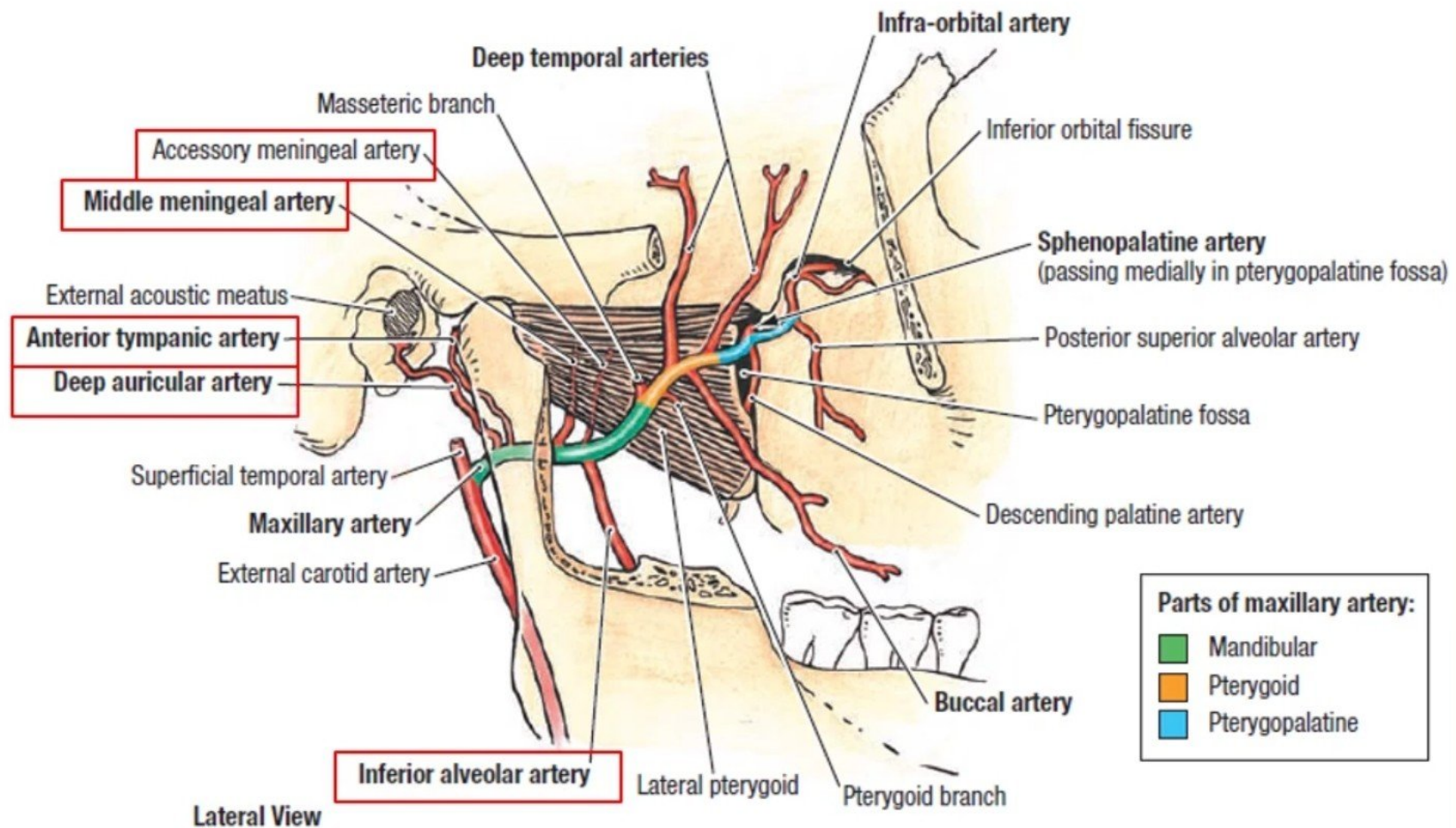
The Maxillary Artery

- ▶ It is the terminal part of the external carotid artery.
- ▶ The maxillary artery is divided into four parts:
 1. Mandibular part.
 2. Muscular part. (pterygoid)
 3. Infraorbital part.
 4. Sphenopalatine part. (pterygopalatine)



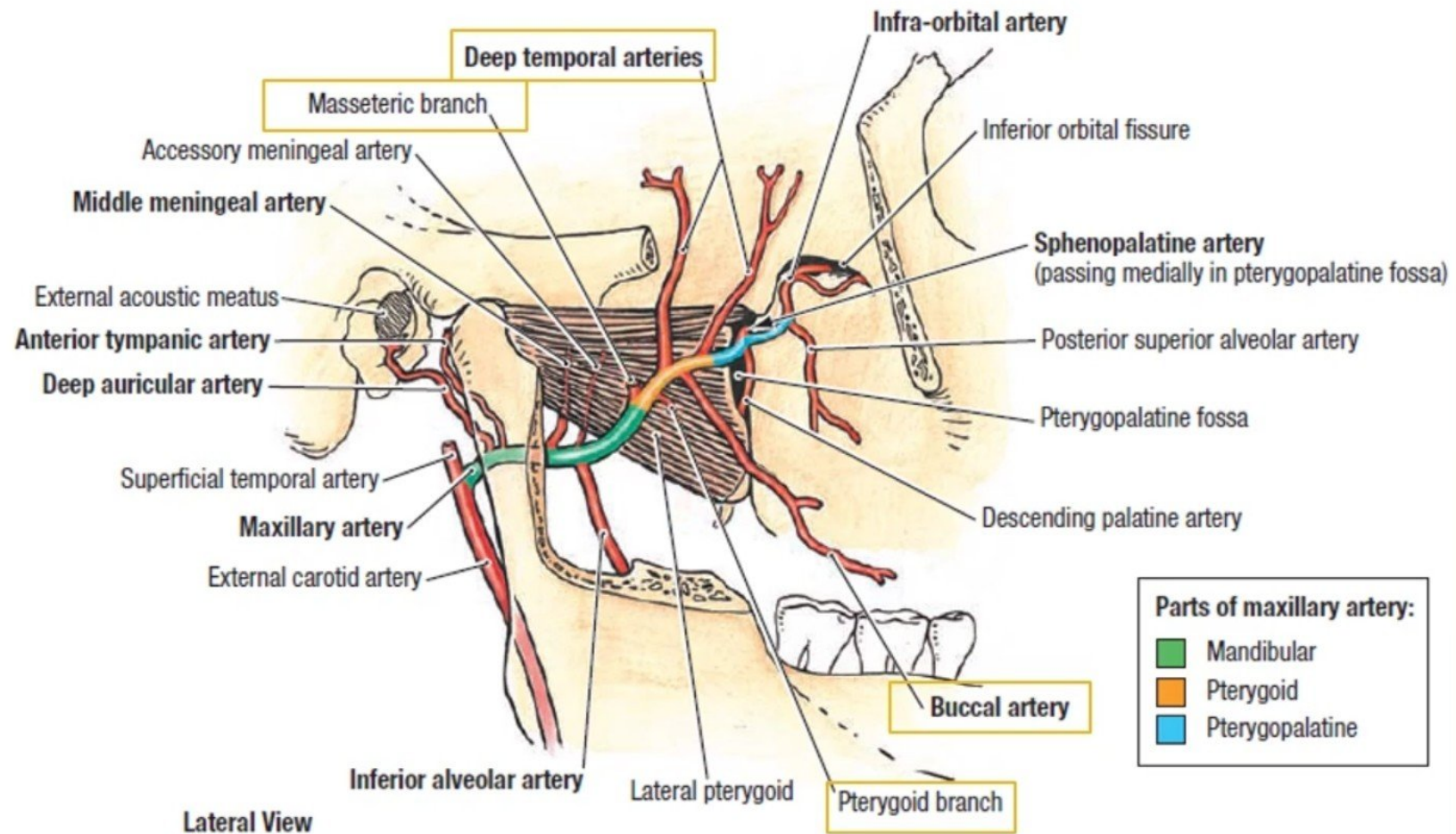
1.The mandibular part of maxillary artery

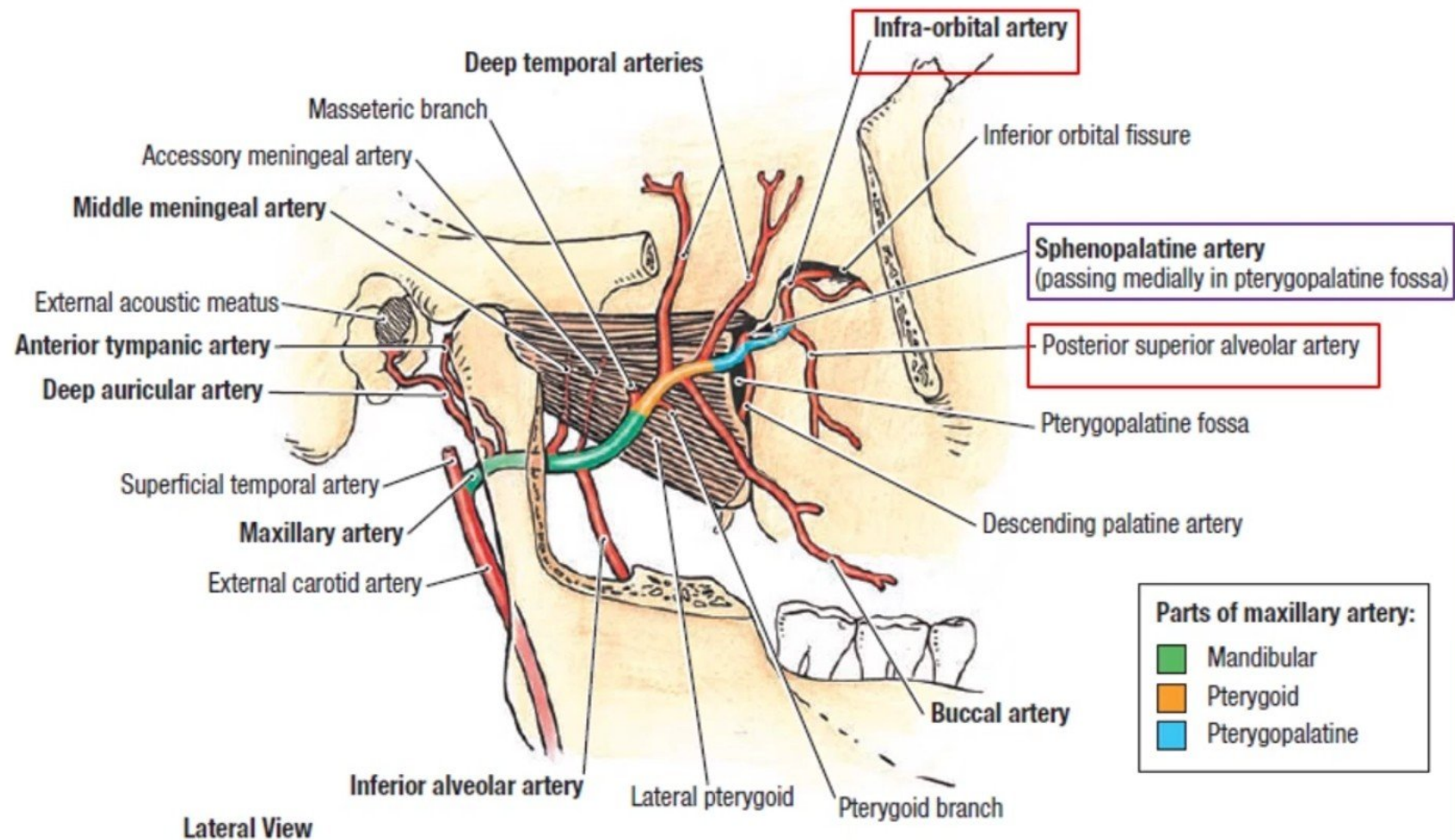
- ▶ Deep auricular and anterior artery.
- ▶ Tympanic arteries.
- ▶ Middle meningeal artery.
- ▶ Accessory meningeal artery.
- ▶ Inferior alveolar artery.



2.The muscular (pterygoid) part of maxillary artery

- ▶ Deep temporal arteries.
- ▶ Massetric and pterygoid arteries.
- ▶ Buccal artery.





3. Infraorbital part

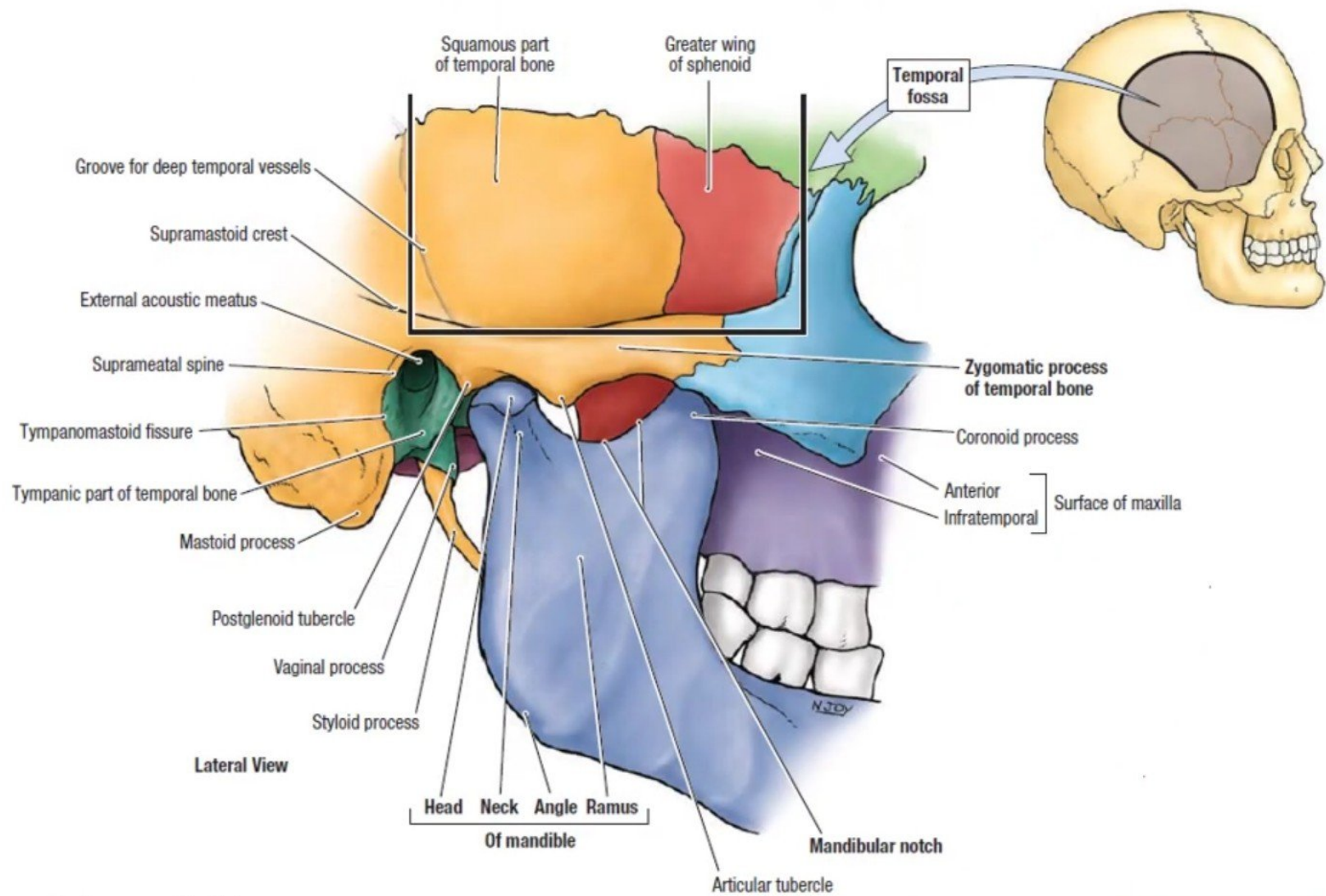
► It give off the:

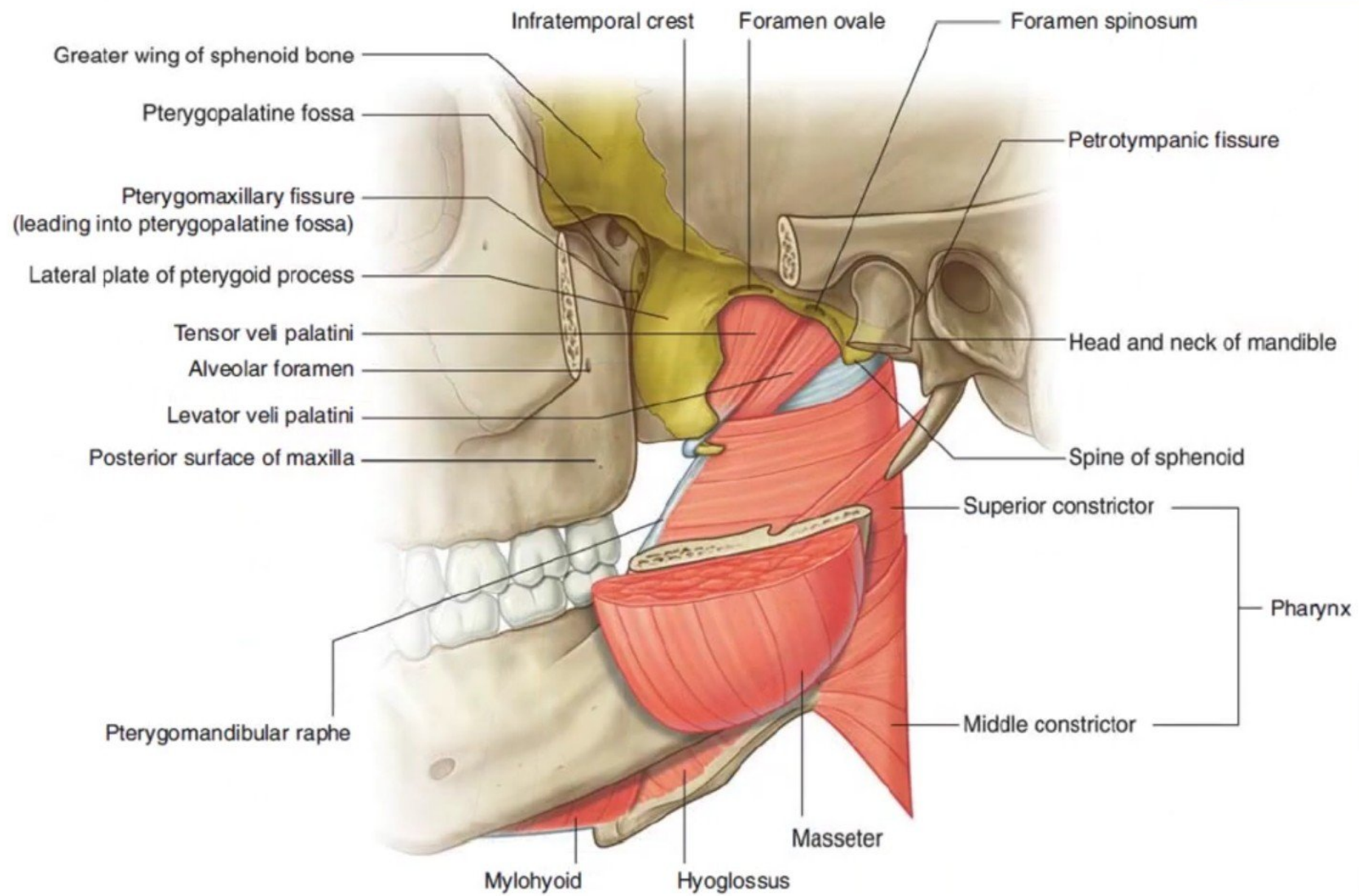
- Superior alveolar arteries.
- Infraorbital artery.

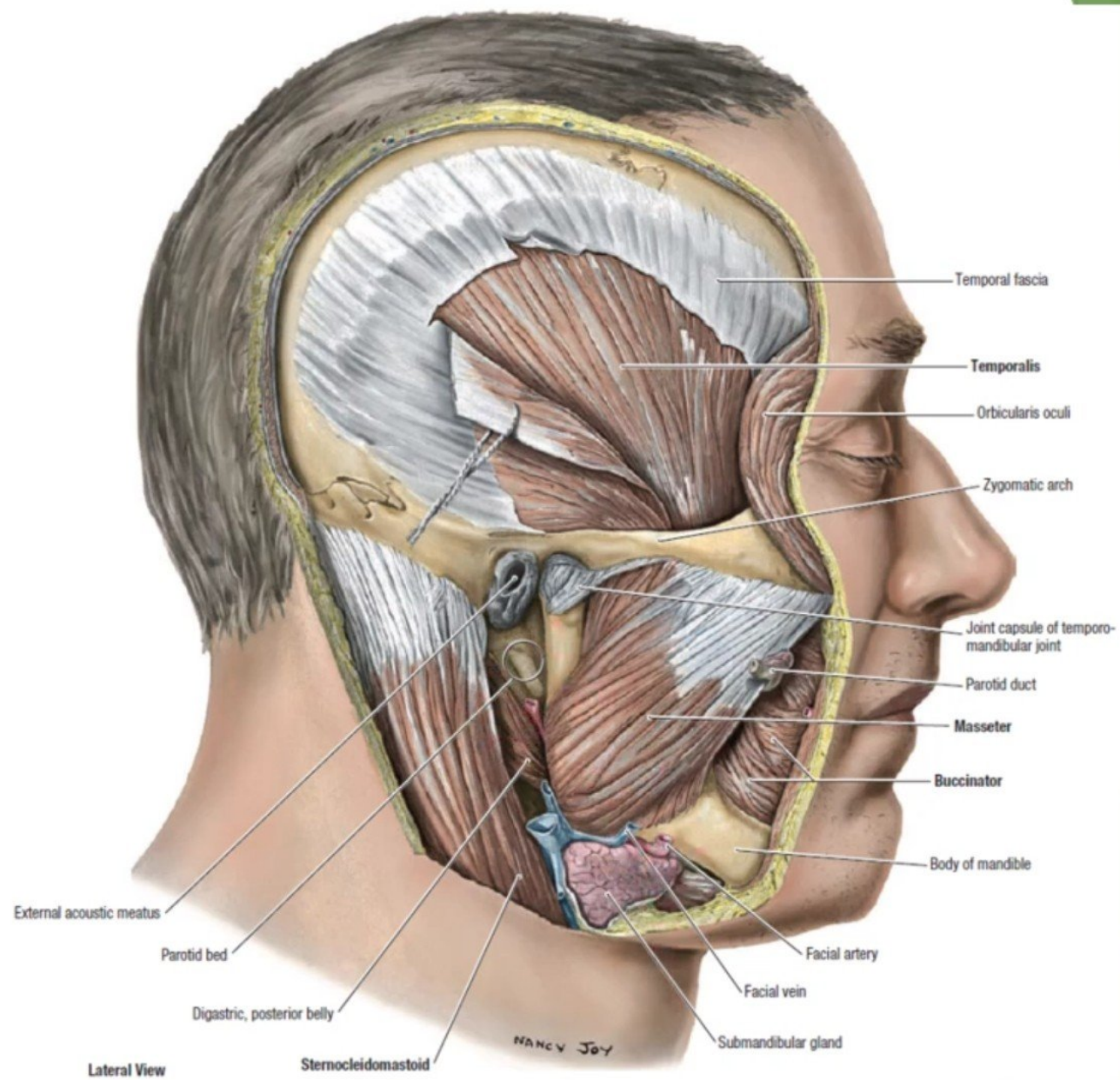
4. Sphenopalatine part of the maxillary artery

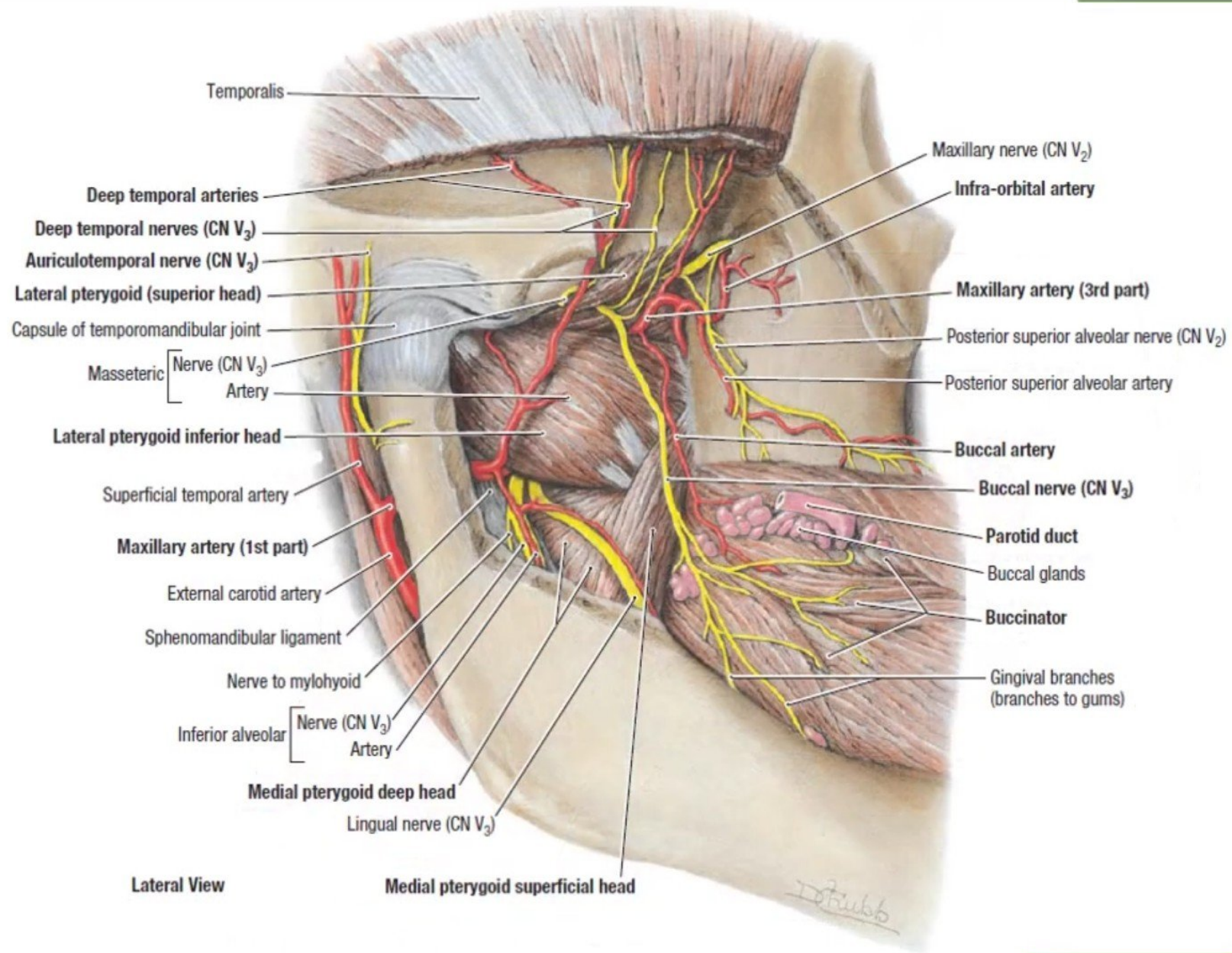
This part gives off the branches which supply

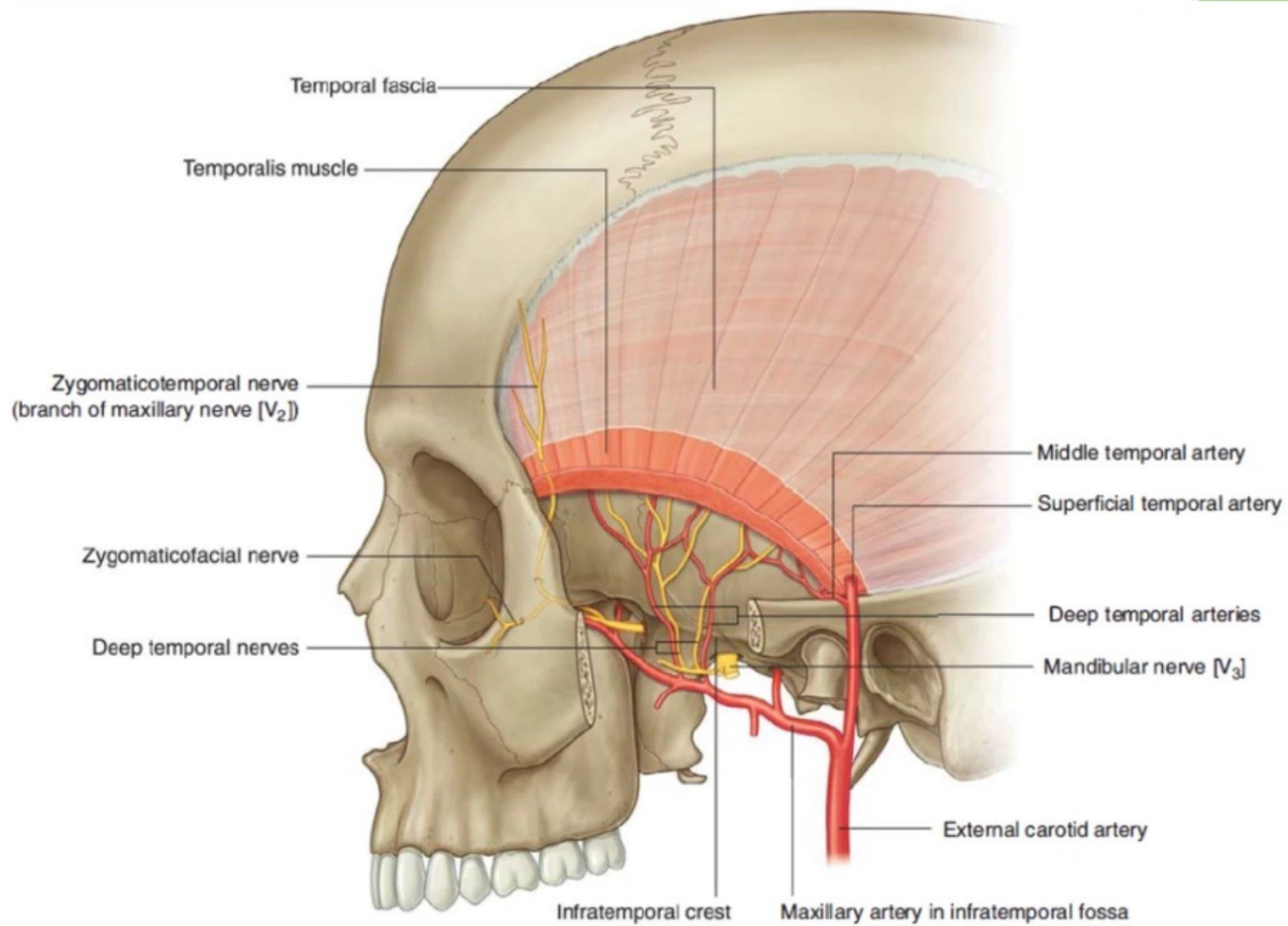
- ❖ The nasal cavity.
- ❖ The palate of oral cavity.





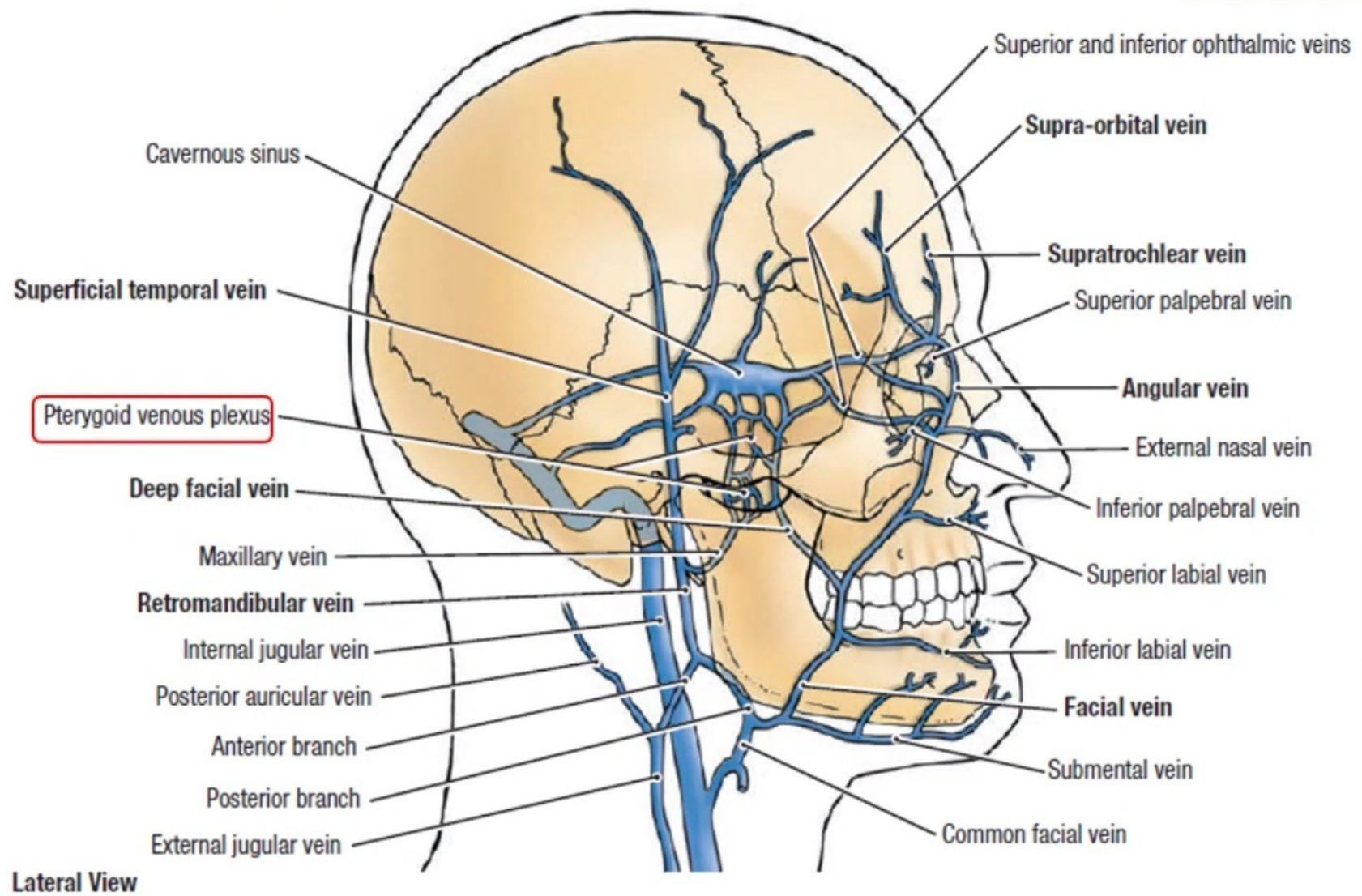


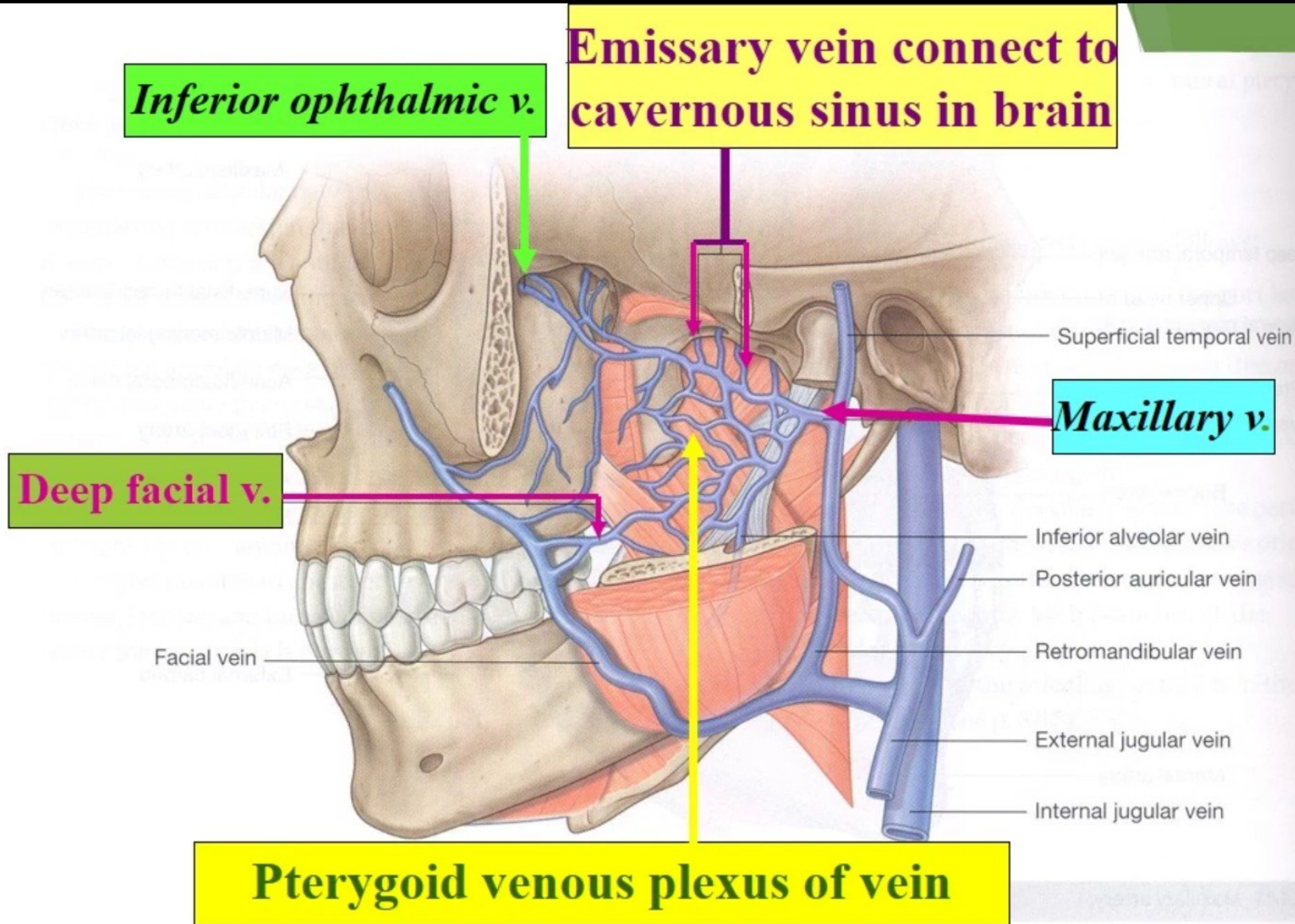


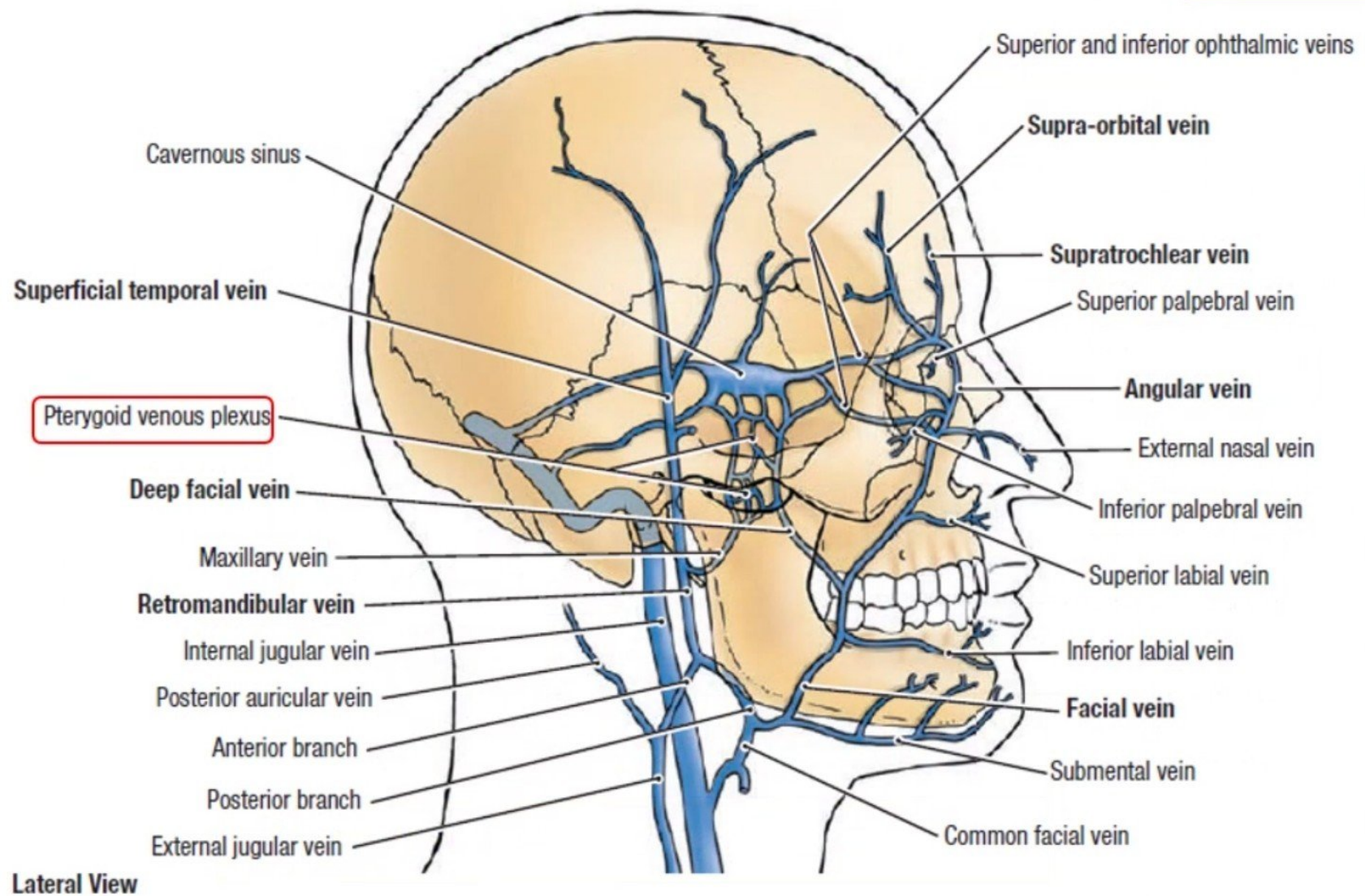


The pterygoid venous plexus

- It is the venous plexus which locate around the pterygoid muscle. Moreover, it has connection to these veins.
 - 1. The maxillary vein.
 - 2. Deep facial vein.
 - 3. Inferior ophthalmic vein.
 - 4. Cavernous sinus (in the cranial cavity).
- *Therefore, the infection could invade from external to internal area.*







Trigeminal nerve

```
graph TD; A[Trigeminal nerve] --> B[Ophthalmic]; A --> C[Maxillary]; A --> D[Mandibular]; B --> E[GSA]; C --> F[GSA]; D --> G[GSA]; D --> H[SVE];
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Ophthalmic

GSA

Maxillary

GSA

Mandibular

GSA

SVE

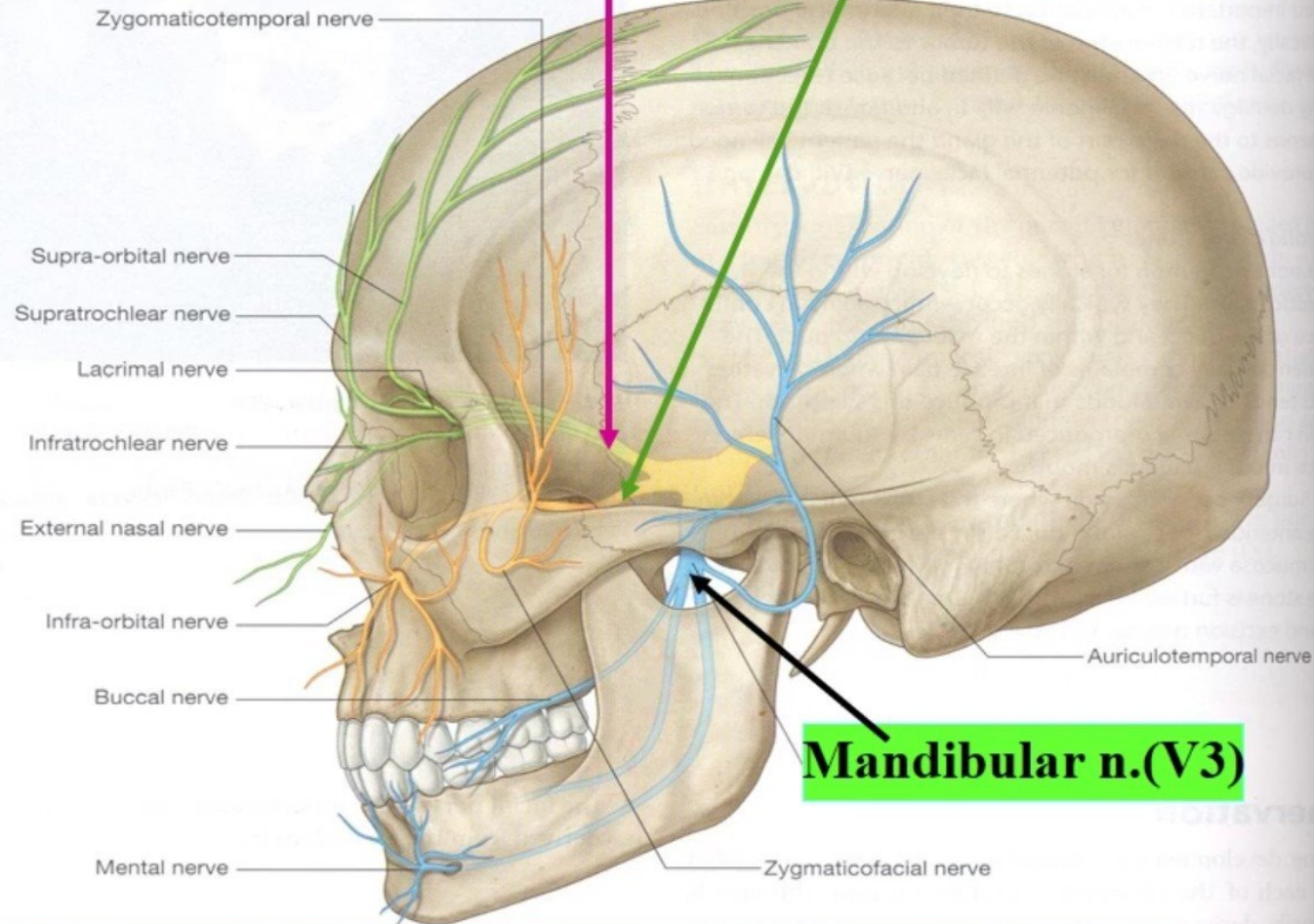
Trigeminal nerve

Components of fibers:

- ▶ **SVE (Special visceral efferent) fibers:**
originate from motor nucleus of trigeminal nerve, and supply masticatory muscles.
- ▶ **GSA (General somatic afferent) fibers:**
transmit facial sensation to sensory nuclei of trigeminal nerve, the GSA fibers have their cell bodies in trigeminal ganglion, which lies on the apex of petrous part of temporal bone.

Ophthalmic n. (V1)

Maxillary n. (V2)



Mandibular n. (V3)

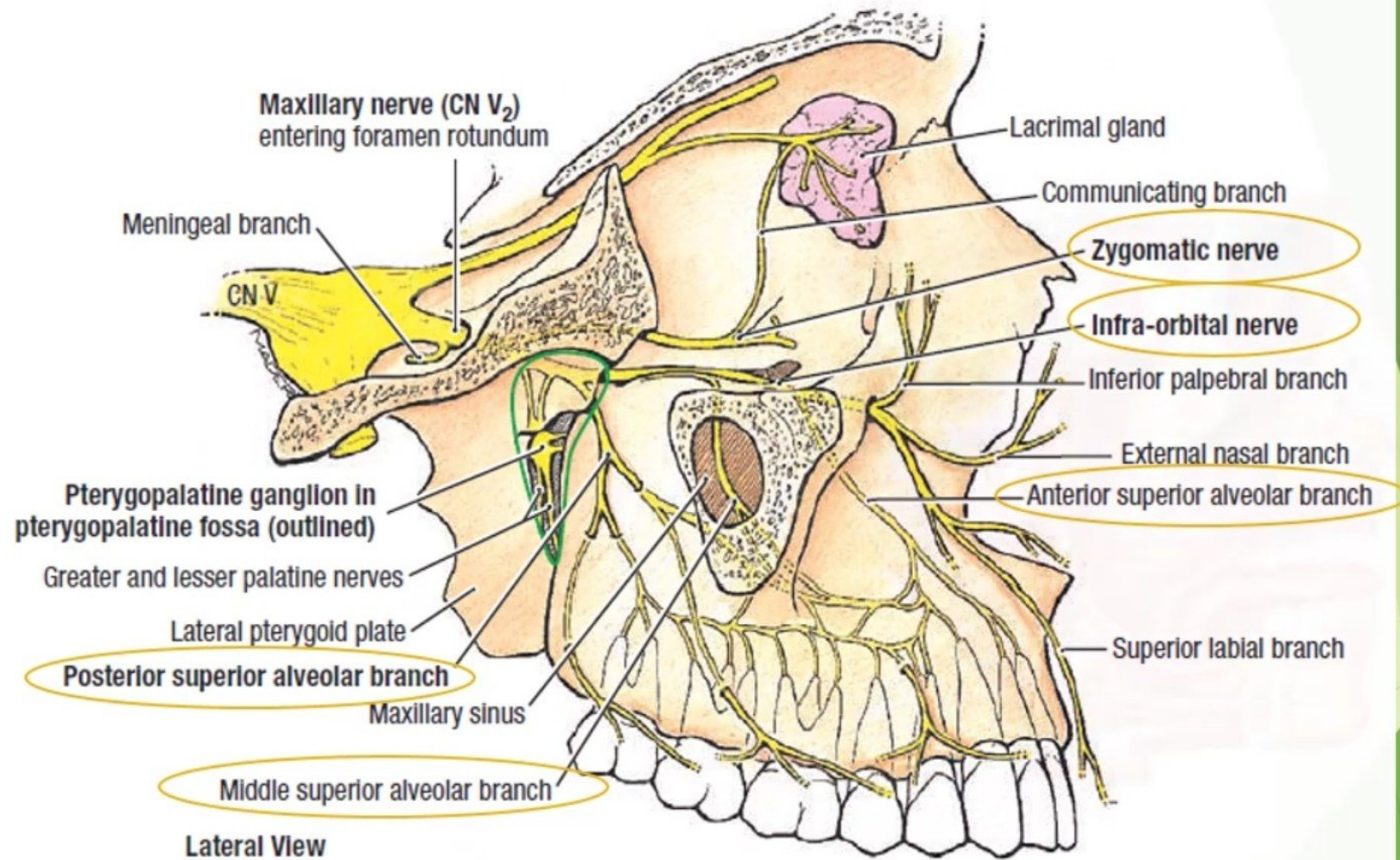
Trigeminal nerve (V) leaving the skull



(A) Lateral view

Maxillary nerve (V2, sensory)

- ▶ Leave skull through foramen rotundum.
- ▶ Branches:
 - ▶ Infraorbital nerve.
 - ▶ Zygomatic nerve.
 - ▶ Superior alveolar nerve.
 - ▶ Pterygopalatine nerve.



Mandibular nerve (V3, mixed)

Leave the skull through the
foramen ovale to enter the
infratemporal fossa

Branches

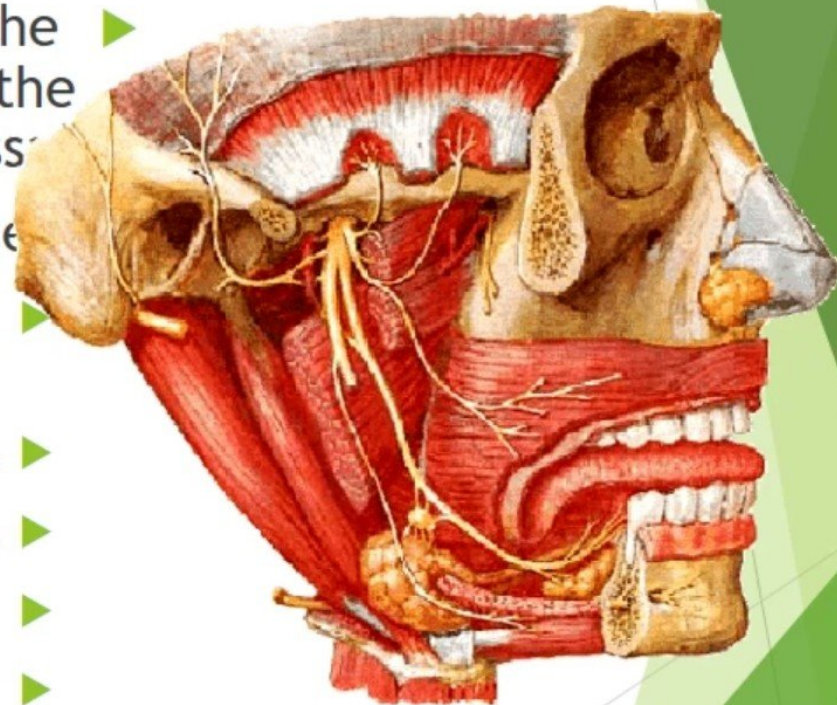
Auriculotemporal
nerve.

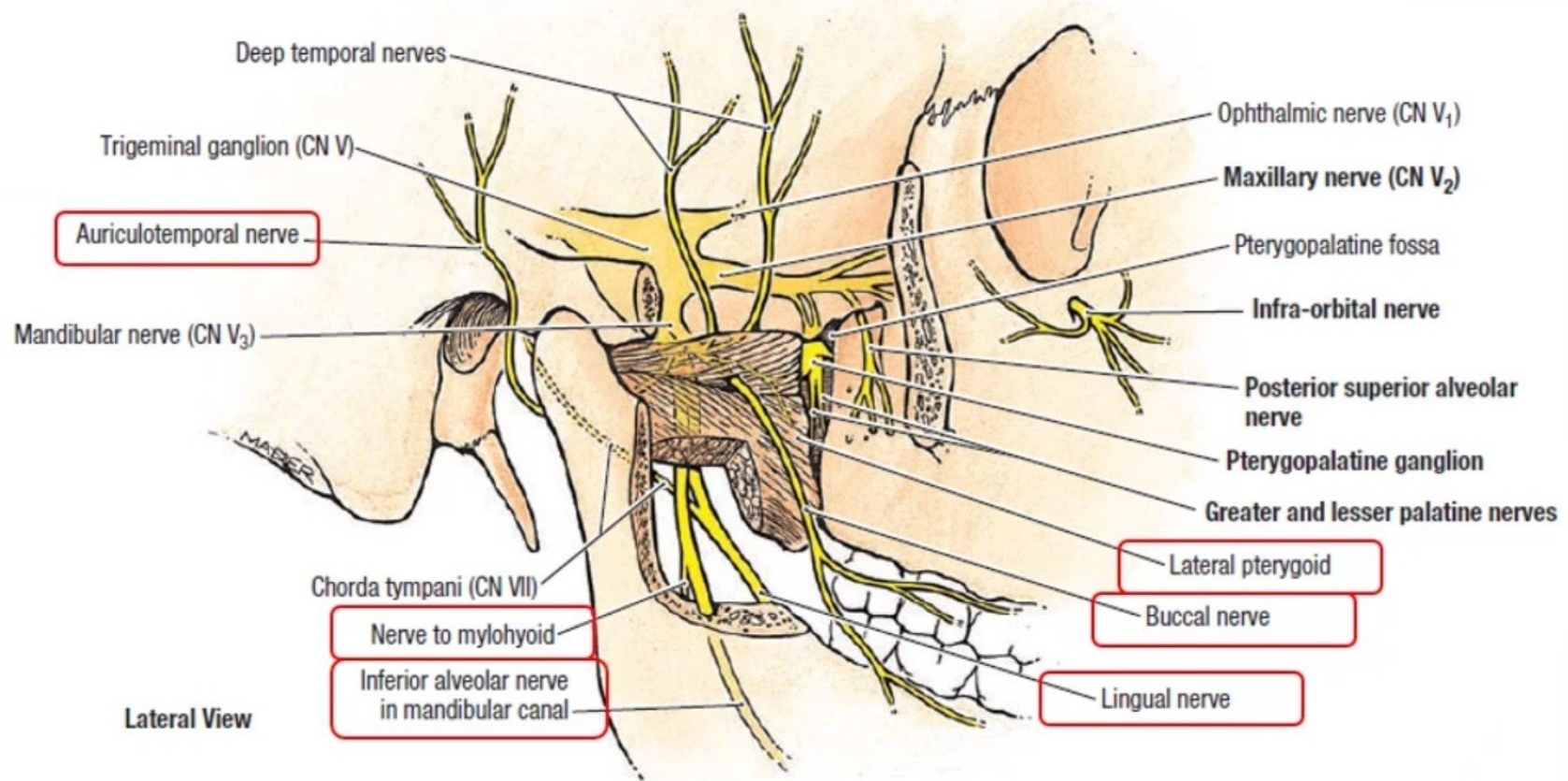
Buccal nerve.

Lingual nerve.

Inferior alveolar nerve.

Nerve of masticatory
muscles.



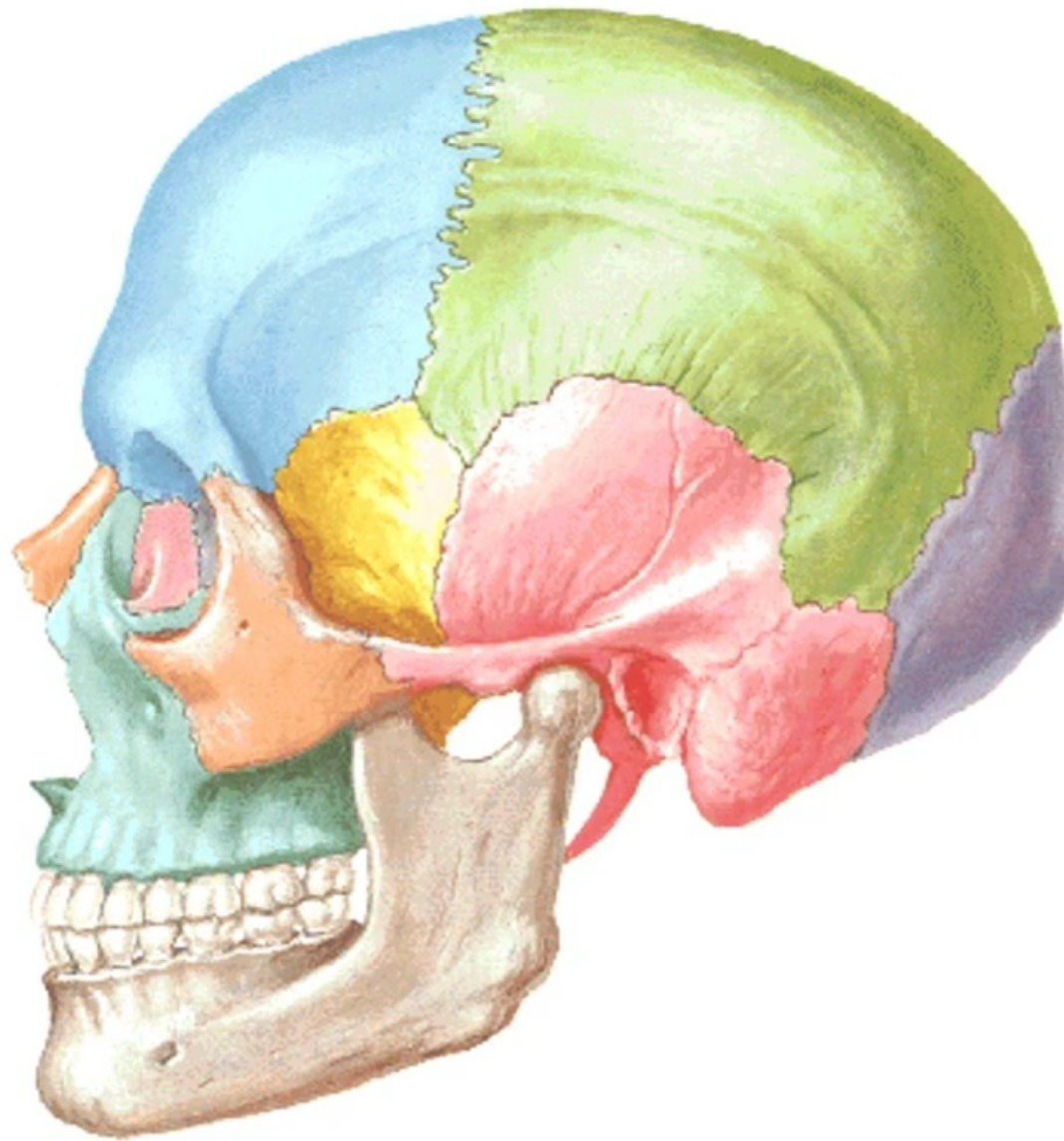


The summery of the temporal and infratemporal fossa

Temporal Fossa

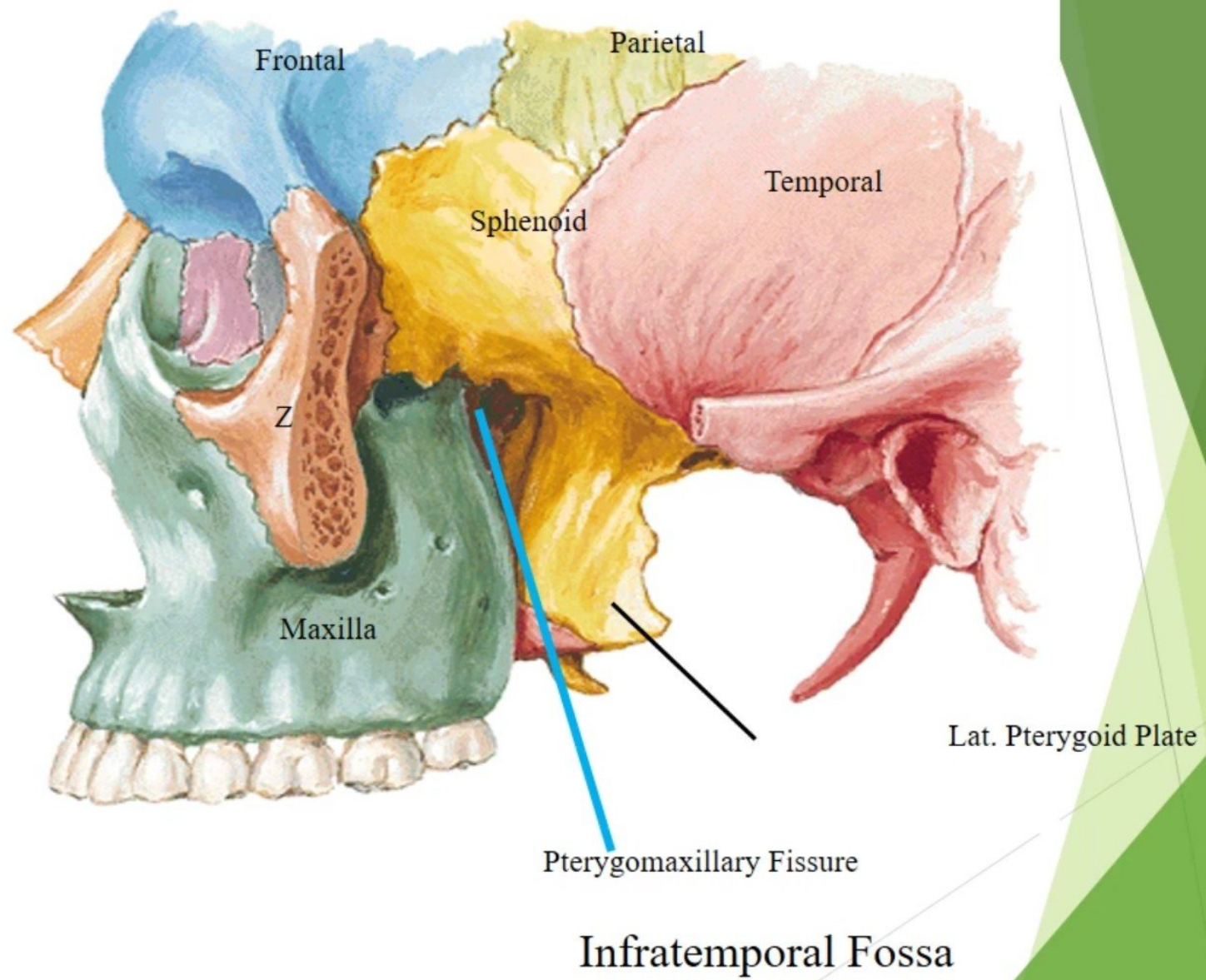
❑ It is boundaries:

- Anterior: zygomatic process of the frontal bone and the frontal process of the zygomatic bone.
- Above (superior): temporal line.
- Below (Inferior): zygomatic arch. (infratemporal crest).
- Lateral wall (roof): the temporal fascia.
- Medial wall (floor): Pterion (Frontal + Parietal + Squamous part of the temporal bone + Greater wing of the sphenoid).



Infratemporal Fossa

- ❑ It is boundaries:
 - **Anterior:** posterior surface of the maxilla. (the inferior orbital fissure).
 - **Posterior:** styloid (carotid sheath) and mastoid processes.
 - **Medial:** lateral pterygoid plate of the sphenoid bone (the tensor palati muscles & superior constrictor of the pharynx).
 - **Lateral:** the ramus and coronoid process of the mandible.
 - **Roof:** greater wing of the sphenoid and infratemporal crest (foramen oval & spinosum).



Temporal Fossa

❑ It contains:

- The temporalis muscle.
- The deep temporal nerves.
- The deep temporal vessels.
- The auriculotemporal nerve.
- The superficial temporal vessels.

Infratemporal Fossa

❑ It contains:

- The deep part of the parotid gland.
- The insertion of lower portion of the temporalis muscle into the coronoid process .
- The lateral and medial pterygoid muscles.
- The pterygoid plexus of veins.
- The mandibular nerve and its branches.
- The maxillary artery and its branches.
- The chorda tympani.
- The otic ganglion.
- The posterior superior alveolar branches of the maxillary nerve.

Thanks